THE SHEAF OF DIFFERENTIALS OF A FORMALLY SMOOTH MORPHISM OF ALGEBRAIC SPACES IS LOCALLY PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4768-4773 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 4775-4800, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally smooth morphisms of algebraic spaces, etale localisation of differentials, and local projectivity of a sheaf of modules; the chart reduction is retained.

Source reference. Original lemma algebra-lemma-characterize-formally-smooth-again: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4800–4800. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-helper-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4554–4565. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-affine-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4800–4800. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma properties-section-locally-projective: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4800–4800. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a formally smooth morphism of algebraic spaces over S. Then $\Omega_{X/Y}$ is locally projective on X.*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Choose a diagram

$$\begin{array}{ccc} U & \xrightarrow{\psi} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are affine(!) schemes and the vertical arrows are étale. By Lemma lemma-helper-formally-smooth we see $\psi : U \to V$ is formally smooth. Hence $\Gamma(V, \mathcal{O}_V) \to \Gamma(U, \mathcal{O}_U)$ is a formally smooth ring map, see More on Morphisms, Lemma more-morphisms-lemma-affine-formally-smooth. Hence by Algebra, Lemma algebra-lemma-characterize-formally-smooth-again the $\Gamma(U, \mathcal{O}_U)$-module $\Omega_{\Gamma(U,\mathcal{O}_U)/\Gamma(V,\mathcal{O}_V)}$ is projective. Hence $\Omega_{U/V}$ is locally projective, see Properties, Section properties-section-locally-projective. Since $\Omega_{X/Y}|_U = \Omega_{U/V}$ we see that $\Omega_{X/Y}$ is locally projective too. (Because we can find an étale covering of X by the affine U's fitting into diagrams as above – details omitted.) $\square$